

Anterior Cruciate Ligament Replacement: Comparison of Bone-Patellar Tendon-Bone Grafts with Two-Strand Hamstring Grafts

A PROSPECTIVE, RANDOMIZED STUDY

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Background: The purpose of this investigation was to evaluate replacement of a torn anterior cruciate ligament with either a bone-patellar tendon-bone autograft or a two-strand semitendinosus-gracilis autograft to compare the results of clinical testing, patient satisfaction, activity level, functional status, and muscle strength.

Methods: Fifty-six patients with a torn anterior cruciate ligament were enrolled in a prospective, randomized, controlled study. Twenty-eight underwent reconstruction with a bone-patellar tendon-bone autograft, and twenty-eight were treated with a two-strand semitendinosus-gracilis autograft. Patients were followed for an average of thirty-nine months (range, thirty-six to fifty-seven months). At the time of final follow-up, twenty-two patients in each group were evaluated in terms of clinical test findings, patient satisfaction, activity level, functional status, and isokinetic muscle strength.

Results: The objective outcome of replacement of the torn anterior cruciate ligament with a bone-patellar tendon-bone graft was superior to that obtained with a two-strand semitendinosus-gracilis graft. At the three-year follow-up interval, the patients in whom a hamstring graft had been used had an average of 4.4 mm of increased anterior knee laxity compared with the laxity of the contralateral, normal knee, whereas the patients in whom a bone-patellar tendon-bone graft had been used had an average of 1.1 mm of increased knee laxity. Fourteen percent (three) of the twenty-two patients with a hamstring graft had a mild pivot shift, and 27% (six) had a moderate pivot shift. Only 14% (three) of the twenty-two patients with a bone-patellar tendon-bone graft had a mild pivot shift, and none had a moderate pivot shift. At the same follow-up interval, the patients in whom a hamstring graft had been used had significantly lower peak knee-flexion strength than those who had a bone-patellar tendon-bone graft ($p = 0.039$). In contrast, the two treatments produced similar outcomes in terms of patient satisfaction, activity level, and knee function (ability to perform a one-legged hop, bear weight, squat, climb stairs, run in place, and duckwalk).

Conclusions: After three years of follow-up, the objective results of anterior cruciate ligament replacement with a bone-patellar tendon-bone autograft were superior to those of replacement with a two-strand semitendinosus-gracilis graft with regard to knee laxity, pivot-shift grade, and strength of the knee flexor muscles. However, the two groups had comparable results in terms of patient satisfaction, activity level, and knee function.

Of all the knee ligaments, the anterior cruciate is totally disrupted the most frequently¹⁻³. Many of these disabling injuries occur in young athletes, predisposing them to degenerative changes of the tibiofemoral joint. These changes

occur because of loss of the primary function of the anterior cruciate ligament, which is to prevent anterior subluxation of the tibia relative to the femur^{4,5}. Following disruption of the anterior cruciate ligament, repeated episodes of joint instability may occur and be associated with meniscal injury, increased erosion of joint cartilage, and abnormal osseous metabolic activity⁶⁻⁹.

The bone-patellar tendon-bone graft is the most popular autogenous graft for intra-articular reconstruction of the



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anterior cruciate ligament, and its popularity and success obligate authors investigating other graft materials to use it as a basis of comparison^{1,10-19}. Variations in the preparation of bone-patellar tendon-bone grafts and in the technique of implantation are described in the literature, with various results^{1,3,10,11,14-18,20-22}. Although several of these studies have demonstrated a high degree of success, it is clear that satisfactory results are not universal^{12,14,21}. An important concern is the fact that the prevalence of anterior knee pain following anterior cruciate ligament reconstruction with a bone-patellar tendon-bone autograft has been reported to be higher than that following reconstruction with a hamstring autograft²³⁻²⁶, although recent evidence suggests that techniques of more accelerated rehabilitation result in a lower prevalence of anterior knee pain²⁷. The hamstring tendons (the semitendinosus and gracilis) are also frequently used as graft tissue to replace the anterior cruciate ligament²⁸⁻³¹, and numerous reports of acceptable results support the use of these tendons in two or four-strand combinations^{10,11,15-18,20,32}. Sacrifice of these tendons seems to be associated with minimal morbidity³³; however, it is unclear whether these tendons stabilize the knee adequately after they have been transplanted as an intra-articular anterior cruciate ligament reconstruction.

After reviewing the literature, Frank and Jackson⁵ concluded that "clinical comparisons have shown that bone-patellar ligament-bone and hamstring grafts have similar rates of effectiveness in adults, with only minor variations in knee stability and muscle strength." Comparison of the rates of complications associated with these two techniques also failed to identify one type of graft as being clearly superior.

The objective of the present investigation was to perform a prospective, randomized, controlled comparison of bone-patellar tendon-bone grafts and two-strand semitendinosus-gracilis grafts for replacement in patients with a complete tear of the anterior cruciate ligament. Comparisons were made over a three-year interval and consisted of clinical testing (with anterior-posterior knee laxity as a primary outcome) and evaluation of patient satisfaction, activity level, function, and isokinetic muscle strength.

Materials and Methods

Patients and Entry Criteria

Between 1990 and 1991, consecutive male and female patients between eighteen and fifty years of age who sought treatment for a tear of the anterior cruciate ligament at our University Sports Medicine Center were asked to participate in the study. Patients were excluded if they had had a previous injury of or operation on either knee or if they had a concurrent fracture, osteoarthritis, or injury of the posterior cruciate ligament, lateral collateral ligament, or posterolateral corner of the knee or a grade-III tear³⁴ of the medial collateral ligament. Patients with a grade-I or II tear of the medial collateral ligament were considered eligible. The investigation was approved by the Institutional Committee on Human Research prior to commencement, and all patients gave informed consent prior to participation.

Fifty-six patients with a mean age of 29.2 years (range, eighteen to forty-six years) provided informed consent and participated. Thirty-one patients were male, and twenty-five were female. The right knee was injured in twenty-four patients and the left, in thirty-two.

With use of a random-number table, the patients were randomized to undergo reconstruction of the anterior cruciate ligament with either a bone-patellar tendon-bone or a two-strand semitendinosus-gracilis autograft. Twenty-eight patients were randomized to each group. Age, gender, activity level, intraoperative findings, and concomitant injuries were similar between the two groups (Table I), indicating that the randomization procedure had been effective in establishing two groups with similar characteristics. Eighty-two percent of the patients in each group had injured the knees during a sports activity. The time-interval between the index injury and the surgery was similar for the two groups (a mean of 109 days with a range of eighteen to 305 days for the group treated with the bone-patellar tendon-bone graft and a mean of seventy-seven days with a range of seventeen to 270 days for the group treated with the semitendinosus-gracilis graft; $p = 0.45$).

The surgery was carried out by one of three surgeons using identical techniques. At the time of arthroscopy, the knee was examined, associated joint pathology was documented, and irreparably torn meniscal fragments were removed. The femoral notch was then enlarged as necessary to improve visualization and to avoid later impingement against the graft. An arthroscopically assisted two-incision technique was used in all knees.

Surgical Technique

Bone-patellar tendon-bone grafting: The bone-patellar tendon-bone graft was constructed from the central third of the tendon of the ipsilateral knee. The graft was 10 mm wide and incorporated in continuity bone blocks (each 20 to 28 mm long) removed from the tibial tubercle and patella. The patellar defect was grafted with cancellous bone trimmed from the bone blocks or with bone curetted from the bed of the tibial donor site. An Acufex (Acufex Microsurgical, Norwood, Massachusetts) drill-guide system was used to place 2.0-mm guide-pins in the center of the tunnels to be reamed in the tibia and femur. The tibial pin was located at the junction of the middle and posterior thirds of the normal site of attachment of the anterior cruciate ligament to the tibia. An Acufex rear-entry guide helped to direct a pin from the lateral aspect of the distal part of the femur (through a separate incision) to the ten-thirty position (for right knees) or the one-thirty position (for left knees) at the back of the intercondylar notch. This pin was placed to create a tunnel in the distal part of the femur that had no more than 1 mm of bone behind the tunnel. The guide-pins were reamed to an appropriate size depending on the width of the autograft bone blocks. The edges of the tunnels within the notch were carefully chamfered.

The bone blocks of the graft were trimmed to appropriate sizes. Three 1.5-mm holes were drilled into each bone block

and number-5 Mersilene sutures were passed through each hole. The graft was pulled through the tunnels so that the patellar bone block was within the femoral tunnel and the tibial bone block was within the tibial tunnel. The graft was positioned so that no bone protruded into the joint. An interference-fit screw was used in the femoral tunnel to fix the bone block. Tension was then placed on the distal part of the graft, and the knee was cycled through a range of motion. If the bone block moved within the tunnel as the knee was flexed and extended, the intercondylar notch was debrided. The knee was then extended, and the graft was pulled in a distal direction. Next, the graft was recessed 1 mm and was fixed at an appropriate tension within the tibial tunnel with use of an interference screw.

Semitendinosus-gracilis grafting: The tunnels for the hamstring graft were placed in exactly the same manner as the tunnels for the bone-patellar tendon-bone graft, but they were then drilled to a smaller size to accommodate the thinner grafts. A skin incision was made over the pes anserinus tendons, and the distal insertion sites of the semitendinosus and gracilis tendons were identified and isolated. An opened tendon stripper was placed over each tendon to detach the muscle. The semitendinosus and gracilis tendons were sutured together and then passed under the sartorius tendon insertion and through the tibial and femoral tunnels. The proximal end of the graft was fixed to the lateral-distal aspect of the femur with use of a belt-buckle-type double-staple technique³⁵. A staple was placed over the hamstring graft, just distal to the tibial tunnel.

Postoperative Rehabilitation Program

The postoperative regimen was identical for the two groups of patients. An adjustable hinged brace locked at 10° of flexion was placed on the knee and was worn continuously for the first week. The patient walked with toe-touch weight-bearing using crutches. On the seventh postoperative day, the brace was adjusted to allow motion between 0° and 70° of flexion, quadriceps activity against gravity was permitted with the knee between 45° and 70° of flexion, hamstring contractions were permitted, and the patient continued to walk with toe-touch weight-bearing using crutches. Three weeks after surgery, the brace was adjusted to allow between 0° and 90° of flexion, and quadriceps exercises were allowed between 45° and 90° of flexion with a maximum of 5 lb (2.3 kg) of weight. At three weeks, the patient was permitted to bear weight as tolerated without crutches while wearing the brace. At five weeks, use of the brace was discontinued except in dangerous situations such as uneven terrain or crowds. The patient was allowed to extend the knee against a resistance of ≤15 lb (≤6.8 kg) of weight between 45° and 90° of flexion. Full isotonic hamstring contraction, hip abductor-adductor exercises, and swimming were permitted. The patient was allowed to ride a stationary bicycle without resistance. At eight weeks, he or she was encouraged to achieve a full range of motion, to extend the knee against unlimited resistance between 45° and 90° of flexion, and to ride a stationary bicycle with resistance. At twelve weeks, unrestricted isotonic quadriceps-strengthening was allowed between 0° and 90° of flexion. The patient was allowed to ride a bicycle outdoors and to jog at half speed. At

TABLE I Baseline Data at the Time of Anterior Cruciate Ligament Replacement

	Bone-Patellar Tendon-Bone Graft	Semitendinosus- Gracilis Graft
Age (yr)		
Mean	28.5	29.9
Range	18-46	19-42
Sex (no.)		
Male	18	13
Female	10	15
Injured limb (no.)		
Right	11	13
Left	17	15
Meniscal tears (no.)		
Medial	5	2
Lateral	8	7
Medial and lateral	2	5
Meniscal surgery (no.)		
Partial medial meniscectomy	4	1
Partial lateral meniscectomy	7	6
Partial medial and lateral meniscectomy	0	1
Medial meniscal repair	1	1
Medial meniscal repair and partial lateral meniscectomy	1	0

TABLE II Lachman Test Data*

	Bone-Patellar Tendon-Bone Graft†	Semitendinosus- Gracilis Graft†	P Value
Preop.			0.22
0	0 (0)	0 (0)	
1+			
2+	2 (7)	4 (14)	
3+	26 (93)	24 (86)	
Total	28 (100)	28 (100)	
1 yr postop.			0.132
0	8 (33)	4 (21)	
1+	13 (54)	9 (47)	
2+	3 (13)	4 (21)	
3+	0 (0)	2 (11)	
Total	24 (100)	19 (100)	
3 yr postop.			0.001
0	10 (45.5)	3 (14)	
1+	10 (45.5)	6 (27)	
2+	2 (9)	8 (36)	
3+	0 (0)	5 (23)	
Total	22 (100)	22 (100)	

*The preoperative test was performed with the patient under anesthesia. A grade of 0 indicated equal laxity between knees; 1+, a mild difference and a hard end point; 2+, a clear difference and a hard end point; and 3+, a clear difference and a soft end point. There was no difference in the findings of the Lachman test between the treatment groups preoperatively or at the one-year interval; however, the patients with a semitendinosus-gracilis graft had a significant increase at three years. †The values are given as the number of patients with the percentage in parentheses.

four months, running in a straight line was allowed and the first isokinetic strength-test was performed. Between six and eight months, the patient was allowed to return to sports if he or she had 90% isokinetic strength compared with that of the contralateral knee, no effusion, and a full range of motion.

Follow-up Evaluations

All knees were examined before surgery; in the operating room immediately after the procedure; and at two, four, six, twelve, and thirty-six months. The primary clinical outcome was anterior-posterior knee laxity, which was measured with a KT-1000 arthrometer (Medmetric, San Diego, California) at 90°-N and 133°-N anteroposteriorly directed loads and with the Lachman test. The secondary clinical outcome measures included the pivot-shift sign (graded as no difference between the knees, a mild difference or glide, a moderate difference or subluxation, or a severe difference or marked subluxation) and the active and passive ranges of knee motion (with the end of the range of motion established by the patient and the examiner, respectively).

Patient satisfaction was evaluated at the thirty-six-month follow-up visit by asking a single question: "How satisfied are you with recovery and return to preinjury activities following your anterior cruciate ligament reconstruction procedure?" Satisfaction was graded ordinally on a scale of 1 (completely unsatisfied) to 10 (completely satisfied). Stiffness of the knee,

catching and locking of the knee, and anterior knee pain were also recorded.

Activity level was recorded with use of the International Knee Documentation Committee (IKDC³⁶) and Tegner scores³⁷ preoperatively and at the one-year and three-year follow-up intervals.

Knee function was assessed with the one-legged-hop test (evaluated preoperatively and at twelve and thirty-six months), the ability to bear weight (graded as full, partial, or impossible at the time of injury, preoperatively, and at two, four, six, twelve, and thirty-six months), difficulty with climbing up and down stairs and ability to run in place (each categorized as either yes or no at the preoperative, twelve-month, and thirty-six-month visits), difficulty with performing the duckwalk (classified as unable to perform, able to perform with pain, or able to perform at the preoperative, twelve-month, and thirty-six-month visits), and difficulty with squatting (assessed as no problem flexing the knees $\geq 90^\circ$, ability to flex the knees $\geq 90^\circ$ with slight difficulty, unable to flex the knees $> 90^\circ$, or unable to squat at the preoperative, four, six, twelve, and thirty-six-month visits).

Thigh muscle strength was measured at the twelve and thirty-six-month follow-up visits with a Cybex isokinetic dynamometer (Lumex, Ronkonkoma, New York) at 60°, 180°, and 240°/sec, allowing determination of peak extension and flexion torque values.

Three-year follow-up was completed for twenty-two (79%) of the patients in each group.

Statistical Methods

Before the investigation was initiated, the sample size was estimated on the basis of the hypothesis that there was no difference in anterior-posterior knee laxity between the treatment groups. A clinically relevant difference between groups was considered to be a 2.5-mm increase in anterior knee laxity compared with the contralateral laxity (a moderate effect size), the associated variation was estimated to be 4.0 mm, and the probability of committing type-II and I errors was chosen as 0.20 and 0.05, respectively. This yielded an estimated sample size of twenty-four patients per group, and when combined with an expected loss-to-follow-up rate of 20% at three years, a sample size of twenty-eight patients per group, or a total of fifty-six patients, was required.

Comparisons between the bone-patellar tendon-bone and semitendinosus-gracilis groups were made with a repeated-measures analysis of variance for the continuous outcome measures (KT-1000 arthrometric measurements, active and passive ranges of knee motion, one-legged hop, patient satisfaction, and strength). The rank sum test was used to compare the pivot-shift and Lachman data, whereas the Pearson chi-

square test was used to compare the presence of stiffness, catching and locking, and anterior knee pain.

Results

Intraoperative Findings and Concomitant Operations

The operative and tourniquet times were significantly less for the patients treated with the semitendinosus-gracilis graft than they were for those treated with the bone-patellar tendon-bone graft ($p < 0.05$): operations in which the bone-patellar tendon-bone graft was used were performed in an average of 2.1 hours and with a tourniquet time of 1.6 hours compared with 1.8 and 1.2 hours, respectively, for those in which the semitendinosus-gracilis graft was used.

The collateral and posterior cruciate ligaments were normal in all patients. There was no difference in the prevalence of meniscal tears between the groups: 54% (fifteen) of the patients in the bone-patellar tendon-bone group had a meniscal tear compared with 50% (fourteen) in the semitendinosus-gracilis group (Table I). There was also no difference in the rate of meniscal surgery (excision or repair) between the groups ($p = 0.8$): 46% (thirteen) of the patients in the bone-patellar tendon-bone group and 32% (nine) of those in the semitendinosus-gracilis group underwent meniscal surgery. In addition, the distribution of damage to the

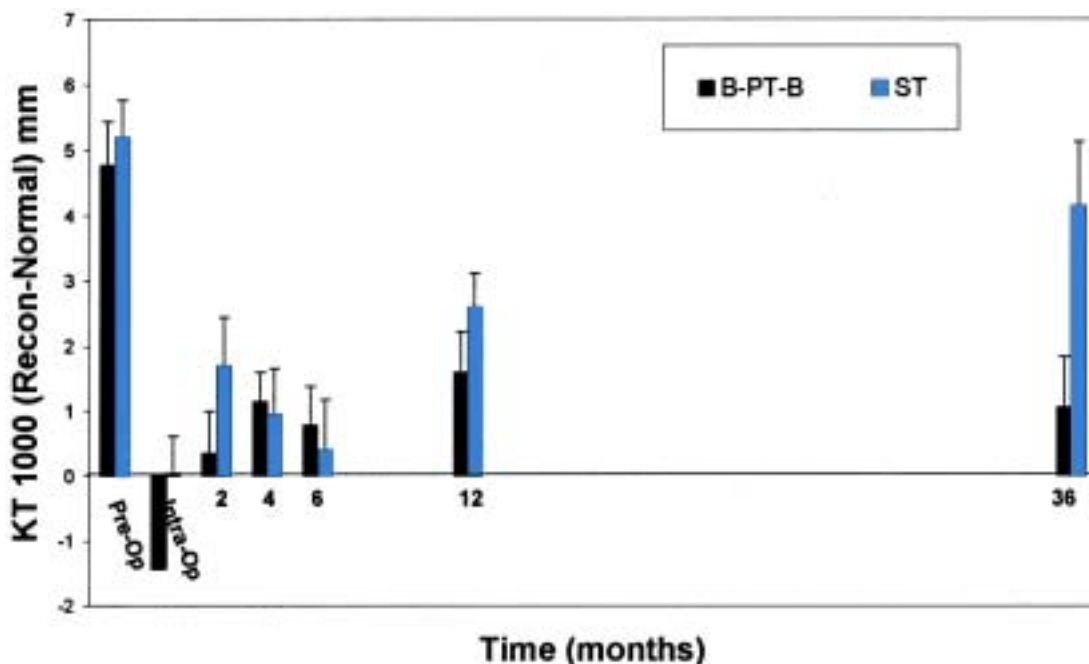


Fig. 1

Anterior-posterior knee laxity (the difference between the reconstructed and normal knees in millimeters) measured with the KT-1000 arthrometer over time. Positive values indicate increased laxity of the reconstructed knee, whereas negative values indicated decreased laxity. There was no difference in laxity values between the bone-patellar tendon-bone (B-PT-B) and semitendinosus-gracilis (ST) groups before the surgery (Pre-Op), just after replacement of the anterior cruciate ligament and with the patient still under anesthesia (Intra-Op), or at one year after the surgery. There was a significant increase in anterior laxity in the semitendinosus-gracilis group compared with the laxity in the bone-patellar tendon-bone group at three years ($p = 0.004$). At that time, 23% (five) of the patients with a bone-patellar tendon-bone graft had >3 mm of increased anterior-posterior laxity compared with that on the contralateral side, and 77% (seventeen) had ≤ 3 mm. At the same time-interval, 55% (twelve) of the patients with a semitendinosus-gracilis graft had >3 mm of increased anterior-posterior laxity compared with that on the contralateral side, and 45% (ten) had ≤ 3 mm.

TABLE III Pivot Shift Data*

	Bone-Patellar Tendon-Bone Graft†	Semitendinosus- Gracilis Graft†	P Value
Preop.			0.936
0	1 (4)	0 (0)	
1+	3 (11)	6 (21)	
2+	11 (39)	9 (32)	
3+	13 (46)	13 (46)	
Total	28 (100)	28 (100)	
1 yr postop.			0.146
0	17 (71)	10 (53)	
1+	6 (25)	5 (26)	
2+	1 (4)	4 (21)	
3+	0 (0)	0 (0)	
Total	24 (100)	19 (100)	
3 yr postop.			0.024
0	19 (86)	13 (59)	
1+	3 (14)	3 (14)	
2+	0 (0)	6 (27)	
3+	0 (0)	0 (0)	
Total	22 (100)	22 (100)	

*The preoperative test was performed with the patient under anesthesia. A grade of 0 indicated no difference between knees; 1+, a mild difference or glide; 2+, a moderate difference or subluxation; and 3+, a severe difference or marked subluxation. There was no difference in the findings of the pivot shift examination between the treatment groups preoperatively or at the one-year interval; however, the patients with a semitendinosus-gracilis graft had a significant increase in grade at three years. †The values are given as the number of patients with the percentage in parentheses.

tibial, femoral, and patellar articular cartilage was equal in the two groups.

Clinical Assessment

Evaluation of anterior-posterior knee laxity with use of the KT-1000 arthrometer at the 133-N load limit revealed similar values for the bone-patellar tendon-bone and semitendinosus-gracilis groups at the preoperative, immediate postoperative, and two, four, six, and twelve-month follow-up intervals (Fig. 1). However, at the three-year follow-up interval, anterior knee laxity in the semitendinosus-gracilis group was significantly greater than that in the bone-patellar tendon-bone group ($p = 0.004$). Immediately postoperatively, the patients with a bone-patellar tendon-bone graft had less anterior knee laxity on the involved side compared with that on the contralateral, normal side ($p = 0.04$), whereas the laxity values for the patients with a semitendinosus-gracilis graft were similar to normal ($p = 0.94$; Fig. 1). At the one and three-year follow-up visits, the patients with a bone-patellar tendon-bone graft had a small increase in anterior knee laxity (average, 1.1 mm at three years) compared with that on the contralateral side, whereas those with a semitendinosus-gracilis graft had a significant increase in anterior laxity (average, 4.4 mm at three years) ($p = 0.001$; Fig. 1). Analysis of the Lachman test data revealed similar findings (Table II), with no difference between the groups at the immediate postoperative or two, four, six, or twelve-month follow-up intervals and then a signifi-

cant increase in anterior laxity in the semitendinosus-gracilis group compared with the laxity in the bone-patellar tendon-bone group at three years ($p = 0.001$). The results of the pivot-shift examination did not differ between the treatment groups immediately postoperatively or at the one-year follow-up visit. However, the percentage of positive pivot-shift signs was significantly higher in the semitendinosus-gracilis group at three years ($p = 0.024$; Table III). At three years, 86% (nineteen) of the patients with a bone-patellar tendon-bone graft had a negative pivot shift sign and 14% (three) had a mild glide. At the same time-interval, 59% (thirteen) of the patients with a semitendinosus-gracilis graft had no pivot shift, three (14%) had a mild glide, and six (27%) had moderate subluxation.

The passive and active ranges of motion were similar between the treatment groups at the one and three-year follow-up intervals. At the three-year follow-up visit, the patients with a bone-patellar tendon-bone graft had, on average, a 19° deficit in active flexion compared with that on the contralateral, normal side, whereas those with a semitendinosus-gracilis graft had an average 12° deficit. At the same follow-up interval, passive flexion was limited by an average of 6° in the bone-patellar tendon-bone group and 7° in the semitendinosus-gracilis group.

Patient Satisfaction

Patients were equally satisfied with the results of both types of treatment. The outcomes were scored as 8.1 ± 1.3 and 8.6 ± 1.2

(average and standard deviation) at the one and three-year follow-up intervals, respectively, in the bone-patellar tendon-bone group and as 7.6 ± 1.8 and 7.9 ± 2.1 in the semitendinosus-gracilis group.

The proportions of patients reporting a feeling of stiffness of the knee were similar between the treatment groups at the one-year follow-up interval. Forty-six percent (eleven) of the twenty-four patients with a bone-patellar tendon-bone graft and 37% (seven) of the nineteen with a semitendinosus-gracilis graft reported a stiff knee at one year ($p = 0.49$). By the three-year follow-up visit, the difference in stiffness approached significance ($p = 0.067$): 55% (twelve) of the twenty-two patients in the bone-patellar tendon-bone group reported a stiff knee, whereas 27% (six) of the twenty-two in the semitendinosus-gracilis group did so.

Reports of catching of the knee were similar between the groups at the one-year and three-year follow-up intervals. At

the three-year follow-up visit, 14% (three) of the twenty-two patients in the bone-patellar tendon-bone group and 9% (two) of the twenty-two in the semitendinosus-gracilis group reported catching.

Anterior knee pain was also similar between the groups at the three-year follow-up visit ($p = 0.44$). Thirty-two percent (seven) of the patients with a bone-patellar tendon-bone graft and 23% (five) of the patients with a semitendinosus-gracilis graft complained of mild anterior knee pain at the three-year follow-up visit.

Activity Level

The activity levels as measured with the Tegner scale were similar between the treatment groups preoperatively and at the one and three-year follow-up intervals (Table IV). Postinjury and preoperatively, the activity level was a median of 4 points for both groups, ranging from 2 to 8 points in the bone-patellar

TABLE IV Tegner Activity Scores and International Knee Documentation Committee (IKDC) Activity Grades*

	Bone-Patellar Tendon-Bone Graft†	Semitendinosus- Gracilis Graft†
Tegner activity score		
Preoperatively		
0-3 points	9 (32)	11 (39)
4-6 points	15 (54)	12 (43)
7-10 points	4 (14)	5 (18)
Total	28 (100)	28 (100)
1 yr postop.		
0-3 points	2 (8)	2 (11)
4-6 points	13 (54)	14 (74)
7-10 points	9 (38)	3 (16)
Total	24 (100)	19 (100)
3 yr postop.		
0-3 points	0 (0)	1 (5)
4-6 points	14 (64)	16 (73)
7-10 points	8 (36)	5 (23)
Total	22 (100)	22 (100)
International Knee Documentation Committee activity grade		
Preoperatively		
IV	1 (4)	1 (4)
III	2 (7)	1 (4)
II	15 (54)	17 (61)
I	10 (36)	9 (32)
Total	28 (100)	28 (100)
3 yr postop.		
IV	0 (0)	0 (0)
III	4 (18)	3 (14)
II	5 (23)	9 (41)
I	13 (59)	10 (45)
Total	22 (100)	22 (100)

*There was no difference in activity levels between the treatment groups at the preoperative, one, or three-year interval. †The values are given as the number of patients with the percentage in parentheses.

TABLE V Extension and Flexion Strength*

	1 Yr Postop.			3 Yr Postop.		
	Bone-Patellar Tendon-Bone Graft†	Semitendinosus- Gracilis Graft†	P Value	Bone-Patellar Tendon-Bone Graft†	Semitendinosus- Gracilis Graft†	P Value
Peak extension torque						
60°/sec	83.5	84.8	0.87	94.7	88.1	0.22
180°/sec	87.7	86.2	0.76	95.9	92.1	0.34
240°/sec	88.3	87.2	0.8	96.6	93.5	0.59
Peak flexion torque						
60°/sec	98.8	98.6	0.97	99.4	95.5	0.37
180°/sec	102.6	95.5	0.11	95.8	90.9	0.23
240°/sec	100.7	95.6	0.15	100.3	89.3	0.039

*As measured with a Cybex isokinetic dynamometer. There was no difference in extension strength values between the treatment groups at the one or three-year interval. However, the patients with the semitendinosus-gracilis graft had a significant deficit in flexion strength at 240°/sec compared with those with the bone-patellar tendon-bone graft at three years. †The values are expressed as a percentage of the torque of the normal knee.

tendon-bone group and from 2 to 10 points in the semitendinosus-gracilis group. These values significantly increased ($p = 0.006$) over the three-year follow-up interval to a median of 6 points (range, 4 to 9 points) in the bone-patellar tendon-bone group and to a median of 5 points (range, 3 to 9 points) in the semitendinosus-gracilis group. The activity grades according to the International Knee Documentation Committee system were similar between the groups at the preoperative and one and three-year follow-up visits (Table IV). The activity grade for both treatment groups improved significantly over time ($p = 0.01$).

Functional Assessment

There was no difference between the treatment groups with regard to the ability to perform the one-legged hop at the one and three-year follow-up intervals. In the bone-patellar tendon-bone group, the differences between the distances that the patients could hop on the uninjured and injured limbs were 12.4 ± 14.9 cm and 7.4 ± 10.7 cm at the one and three-year follow-up visits, respectively; in the semitendinosus-gracilis group, the values were 18.8 ± 11.5 cm and 10.6 ± 9.9 cm, respectively.

All patients walked bearing full weight by the two-month follow-up visit. At each subsequent follow-up interval, there was no difference between the treatment groups with regard to the patient's ability to bear weight, climb stairs, run in place, perform the duckwalk, or squat. At the one-year follow-up visit, 88% (twenty-one) of the twenty-four patients in the bone-patellar tendon-bone group were capable of squatting normally and 13% (three) others could do so with slight difficulty. At the three-year follow-up visit, the corresponding values were 91% (twenty) of twenty-two and 9% (two) of twenty-two. At the one-year follow-up visit, 68% (thirteen) of the nineteen patients in the semitendinosus-gracilis group could squat normally, 26% (five) could do so with slight difficulty, and 5% (one) could not squat past 90° of flexion. At three years, 95% (twenty-one) of the twenty-

two patients could squat normally and one (5%) could do so with slight difficulty.

Muscle Strength

At the one-year follow-up interval, there was no difference in isokinetic muscle strength between the patients with a bone-patellar tendon-bone graft and those with a semitendinosus-gracilis graft (Table V). However, at the three-year follow-up interval, the patients with a semitendinosus-gracilis graft had an 11% decrease in peak flexion torque when the strength evaluation was performed at 240°/sec ($p = 0.039$ for the difference between treatment groups). At the same follow-up interval, there was no deficit in peak extension or flexion torque in the patients with a bone-patellar tendon-bone graft.

Discussion

If one considers this investigation from the perspective of clinical assessment alone, it is clear that the outcome of anterior cruciate ligament replacement with a bone-patellar tendon-bone graft was superior to that of replacement with a two-strand semitendinosus-gracilis graft. On the other hand, the two treatments produced similar outcomes with regard to patient satisfaction, activity level, and knee function. The former statement is substantiated by dramatic increases in anterior knee laxity (as assessed with the KT-1000 arthrometer and Lachman examinations) and pivot-shift grades in the semitendinosus-gracilis group at the three-year follow-up interval compared with relatively unchanged laxity values in the bone-patellar tendon-bone group over the same time-period. We consider these increases in anterior knee laxity to be of substantial clinical and biomechanical concern because they were considerably greater than 3 mm—i.e., the 95% confidence interval based on side-to-side differences in laxity measured in subjects with normal knees. However, it is important for us to point out that the superior laxity values in the bone-patellar tendon-bone group may have come at a cost. At the

three-year follow-up visit, a substantial proportion (55%) of the patients with a bone-patellar tendon-bone graft reported a stiff knee whereas this symptom occurred less frequently among those in the semitendinosus-gracilis group (27% prevalence). This did not appear to have a substantial impact on patient function because both treatments resulted in similar outcomes with regard to active and passive range of knee motion, prevalence of knee-catching, anterior knee pain, and ability to perform the one-legged hop, bear weight, squat, climb stairs, run in place, and duckwalk.

It is unclear why anterior laxity and the pivot shift grades were similar between the groups and within the limits of normal at one year and then increased in the semitendinosus-gracilis group at three years. Studies of the biology of healing anterior cruciate ligament grafts in animal models have suggested that graft-remodeling is complete within the first year³⁸. However, to our knowledge, the time course over which the transplanted graft material is replaced with remodeled tissue that is viable from a biomechanical perspective is unclear, and no investigation that we know of has included three-year data. Consequently, it remains unclear whether the increased anterior knee laxity that occurred between the one and three-year follow-up intervals in our study was due to the fact that the strains applied to the graft and graft fixation were in excess of their yield point during healing or after healing was complete.

Several investigators have recently performed histologic and biomechanical studies concerning healing of hamstring tendons to bone^{20,39-45}. Although there appears to be healing directly to bone, Nagano et al.⁴² demonstrated that the bone blocks of the bone-patellar tendon-bone grafts heal more rapidly to host bone than do hamstring tendons. Rodeo et al.⁴⁴, Blickenstaff et al.³⁹, and Scranton et al.⁴⁵ suggested that care should be taken to avoid overload of hamstring grafts during healing. Thus, on the basis of the available literature, it appears reasonable for the rehabilitation after hamstring grafting to be more conservative than that after bone-patellar tendon-bone grafting. Even though our patients who had hamstring grafting did not have the dramatic increase in anterior knee laxity until after the first year, it is possible that the conservative rehabilitation program that was used for both groups adversely affected the results of the hamstring reconstructions. Alternatively, it is also possible that the more aggressive rehabilitation programs currently being used after replacement of the anterior cruciate ligament with a hamstring graft would have resulted in even greater anterior laxity at an earlier time than we observed^{19,20}.

At the present time, eight prospective studies focusing on the graft material used to replace the anterior cruciate ligament have been published^{10,11,15-18,20,32}. Bone-patellar tendon-bone autografts were compared with four-strand hamstring autografts in seven of these studies^{10,11,15,16,18,20,32}, and they were compared with two-strand hamstring autografts, as they were in our investigation, in one¹⁷. In a prospective, randomized, controlled study of patients with an acute or chronic tear of the anterior cruciate ligament followed for a mean of forty-

two months (range, twenty-four to sixty months), O'Neill¹⁷ compared the results of treatment with a two-strand semitendinosus-gracilis autograft, a central-one-third bone-patellar tendon-bone graft placed with an arthroscopically assisted two-incision technique, and the same graft placed with a single-incision (endoscopic) technique. Although, there were no differences with regard to the patients' symptoms, range of motion, or amount of patellar crepitus, those treated with the bone-patellar tendon-bone graft returned to a higher level of activity. Also, compared with the other two groups, a significantly larger percentage of these patients had a side-to-side difference in anterior knee laxity values of ≤ 3 mm ($p < 0.08$). Of the eight aforementioned prospective studies comparing bone-patellar tendon-bone and hamstring grafts for anterior cruciate ligament replacement, five included measurement of thigh muscle strength^{10,11,16,17,32}. Review of these studies revealed that replacement of the anterior cruciate ligament with an autogenous bone-patellar tendon-bone graft and replacement with a hamstring graft affect the extensor mechanism equally with regard to generation of extension torque, whereas replacement with the hamstring graft appears to adversely affect the ability to develop flexion torque. In the current study, the average peak flexion torque value for the patients who had received the hamstring graft was approximately 10% less than that for the patients who had received the bone-patellar tendon-bone graft. This was noted to occur at an isolated knee angle while the patient was non-weight-bearing; therefore, it is unclear whether a deficit of this magnitude predisposes a patient to increased risk of reinjury.

The semitendinosus-gracilis graft technique used in this study has to be closely scrutinized. Although the strength of a two-strand semitendinosus-gracilis graft is similar to that of an anterior cruciate ligament⁴⁶, neither the strength nor the stiffness is the same as that of the four-strand hamstring graft that most surgeons use today. In addition, the extra-articular fixation that was used in our study was much more compliant than the current fixation, which includes extra-articular and intraosseous components of four-strand hamstring grafts. It is important for us to mention that, although anterior knee laxity was substantially increased in the semitendinosus-gracilis group at the three-year follow-up interval, it is unclear whether such an increase occurs in the four-strand semitendinosus grafts with intraosseous fixation that are popular today. Of the eight prospective studies comparing bone-patellar tendon-bone and hamstring grafts, four showed the graft materials to have similar laxity values^{10,16,20,32} and four showed significantly better values for the bone-patellar tendon-bone grafts^{11,15,17,18}. We can say that two-strand grafts combined with extra-articular fixation are associated with a major problem of an unwanted increase of anterior knee laxity and pivot shift grade, even though our patients appeared to be equally satisfied with both procedures, but we cannot state with certainty that the same increase would occur with the four-strand grafts now being utilized. The increased anterior laxity that occurs with time remains a concern to surgeons using the hamstring graft. Certainly, the frequent reports of new fixation techniques in the literature suggest that the final

solution to fixation of hamstring grafts to bone has not yet been found, and the fixation may have contributed to our findings. Future investigations of anterior cruciate ligament replacement should compare bone-patellar tendon-bone grafts and four-strand grafts with intraosseous fixation combined with accelerated rehabilitation. ■

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References

1. **Eriksson E.** Reconstruction of the anterior cruciate ligament. *Orthop Clin North Am.* 1976;7:167-79.
2. **Feagin JA Jr, Curl WW.** Isolated tear of the anterior cruciate ligament: 5-year follow-up study. *Am J Sports Med.* 1976;4:95-100.
3. **Johnson RJ, Beynnon BD, Nichols CE, Renstrom PA.** Current concepts review. The treatment of injuries of the anterior cruciate ligament. *J Bone Joint Surg Am.* 1992;74:140-51.
4. **Butler DL, Noyes FR, Grood ES.** Ligamentous restraints to anterior-posterior drawer in the human knee. A biomechanical study. *J Bone Joint Surg Am.* 1980;62:259-70.
5. **Frank CB, Jackson DW.** Current concepts review. The science of reconstruction of the anterior cruciate ligament. *J Bone Joint Surg Am.* 1997;79:1556-76.
6. **Andersson C, Odensten M, Gillquist J.** Knee function after surgical or nonsurgical treatment of acute rupture of the anterior cruciate ligament: a randomized study with a long-term follow-up period. *Clin Orthop.* 1991;264:255-63.
7. **Dye SF, Chew MH.** Instructional Course Lecture, the American Academy of Orthopaedic Surgeons. The use of scintigraphy to detect increased osseous metabolic activity about the knee. *J Bone Joint Surg Am.* 1993;75:1388-406.
8. **Dye SF, Wojtyk EM, Fu FH, Fithian DC, Gillquist J.** Instructional Course Lecture, the American Academy of Orthopaedic Surgeons. Factors contributing to function of the knee joint after injury or reconstruction of the anterior cruciate ligament. *J Bone Joint Surg Am.* 1998;80:1380-93.
9. **Liljedahl S-O, Nordstrand A.** Injuries to the ligaments of the knee. Diagnosis and results of operation. *Injury.* 1969;1:17-24.
10. **Aglietti P, Buzzi R, Zaccherotti G, De Biase P.** Patellar tendon versus doubled semitendinosus and gracilis tendons for anterior cruciate ligament reconstruction. *Am J Sports Med.* 1994;22:211-8.
11. **Aglietti P, Zaccherotti G, Buzzi R, De Biase P.** A comparison between patellar tendon and doubled semitendinosus/gracilis tendon for anterior cruciate ligament reconstruction. A minimum five-year followup. *J Sports Traumatol Rel Res.* 1997;19:57-68.
12. **Bach BR Jr, Tradonsky S, Bojchuk J, Levy ME, Bush-Joseph CA, Khan NH.** Arthroscopically assisted anterior cruciate ligament reconstruction using patellar tendon autograft. Five- to nine-year follow-up evaluation. *Am J Sports Med.* 1998;26:20-9.
13. **Buss DD, Warren RF, Wickiewicz TL, Galinat BJ, Panariello R.** Arthroscopically assisted reconstruction of the anterior cruciate ligament with use of autogenous patellar-ligament grafts. Results after twenty-four to forty-two months. *J Bone Joint Surg Am.* 1993;75:1346-55.
14. **Clancy WG Jr, Nelson DA, Reider B, Narechania RG.** Anterior cruciate ligament reconstruction using one-third of the patellar ligament, augmented by extra-articular tendon transfers. *J Bone Joint Surg Am.* 1982;64:352-9.
15. **Corry IS, Webb JM, Clingeleffer AJ, Pinczewski LA.** Arthroscopic reconstruction of the anterior cruciate ligament. A comparison of patellar tendon autograft and four-strand hamstring tendon autograft. *Am J Sports Med.* 1999;27:444-54.
16. **Marder RA, Raskind JR, Carroll M.** Prospective evaluation of arthroscopically assisted anterior cruciate ligament reconstruction. Patellar tendon versus semitendinosus and gracilis tendons. *Am J Sports Med.* 1991;19:478-84.
17. **O'Neill DB.** Arthroscopically assisted reconstruction of the anterior cruciate ligament. A prospective randomized analysis of three techniques. *J Bone Joint Surg Am.* 1996;78:803-13.
18. **Otero AL, Hutcheson L.** A comparison of the doubled semitendinosus/gracilis and central third of the patellar tendon autografts in arthroscopic anterior cruciate ligament reconstruction. *Arthroscopy.* 1993;9:143-8.
19. **Shelbourne KD, Klootwyk TE, Wilckens JH, De Carlo MS.** Ligament stability two to six years after anterior cruciate ligament reconstruction with autogenous patellar tendon graft and participation in accelerated rehabilitation program. *Am J Sports Med.* 1995;23:575-9.
20. **Eriksson K, Anderberg P, Hamberg P, Lofgren AC, Bredenberg M, Westman I, Wredmark T.** A comparison of quadruple semitendinosus and patellar tendon grafts in reconstruction of the anterior cruciate ligament. *J Bone Joint Surg Br.* 2001;83:348-54.
21. **Howe JG, Johnson RJ, Kaplan MJ, Fleming B, Jarvinen M.** Anterior cruciate ligament reconstruction using quadriceps patellar tendon graft. Part I. Long-term followup. *Am J Sports Med.* 1991;19:447-57.
22. **Jones KG.** Reconstruction of the anterior cruciate ligament using the central one-third of the patellar ligament. A follow-up report. *J Bone Joint Surg Am.* 1970;52:1302-8.
23. **Aglietti P, Buzzi R, D'Andria S, Zaccherotti G.** Patellofemoral problems after intraarticular anterior cruciate ligament reconstruction. *Clin Orthop.* 1993;288:195-204.
24. **Kartus J, Magnusson L, Stener S, Brandsson S, Eriksson BI, Karlsson J.** Complications following arthroscopic anterior cruciate ligament reconstruction. A 2-5-year follow-up of 604 patients with special emphasis on anterior knee pain. *Knee Surg Sports Traumatol Arthrosc.* 1999;7:2-8.
25. **Rosenberg TD, Franklin JL, Baldwin GN, Nelson KA.** Extensor mechanism function after patellar tendon graft harvest for anterior cruciate ligament reconstruction. *Am J Sports Med.* 1992;20:519-26.
26. **Sachs RA, Daniel DM, Stone ML, Garfein RF.** Patellofemoral problems after anterior cruciate ligament reconstruction. *Am J Sports Med.* 1989;17:760-5.
27. **Shelbourne KD, Trumper RV.** Preventing anterior knee pain after anterior cruciate ligament reconstruction. *Am J Sports Med.* 1997;25:41-7.
28. **Grana WA, Hines R.** Arthroscopic-assisted semitendinosus reconstruction of the anterior cruciate ligament. *Am J Knee Surg.* 1992;5:16-22.
29. **Karlson JA, Steiner ME, Brown CH, Johnston J.** Anterior cruciate ligament reconstruction using gracilis and semitendinosus tendons. Comparison of through-the-condyle and over-the-top graft placements. *Am J Sports Med.* 1994;22:659-66.
30. **Siegel MG, Barber-Westin SD.** Arthroscopic-assisted outpatient anterior cruciate ligament reconstruction using the semitendinosus and gracilis tendons. *Arthroscopy.* 1998;14:268-77.

31. **Sgaglione NA, Del Pizzo W, Fox JM, Friedman MJ.** Arthroscopically assisted anterior cruciate ligament reconstruction with the pes anserine tendons. Comparison of results in acute and chronic ligament deficiency. *Am J Sports Med.* 1993;21:249-56.
32. **Beard DJ, Anderson JL, Davies S, Price AJ, Dodd CA.** Hamstrings vs. patella tendon for anterior cruciate ligament reconstruction: a randomised controlled trial. *Knee.* 2001;8:45-50.
33. **Yasuda K, Tsujino J, Ohkoshi Y, Tanabe Y, Kaneda K.** Graft site morbidity with autogenous semitendinosus and gracilis tendons. *Am J Sports Med.* 1995;23:706-14.
34. **American Medical Association, Subcommittee on Classification of Sports Injuries and Committee on the Medical Aspects of Sports.** Standard nomenclature of athletic injuries. Chicago: American Medical Association; 1968. p 99-100.
35. **Beynnon BD, Uh BS, Pyne JI, Gottlieb DJ, Johnson RJ.** Semitendinosus and gracilis tendon graft fixation for ACL reconstructions. *Iowa Orthop J.* 1996; 16:118-21.
36. **Hefti F, Muller W, Jakob RP, Staubli HU.** Evaluation of knee ligament injuries with the IKDC form. *Knee Surg Sports Traumatol Arthrosc.* 1993;1:226-34.
37. **Tegner Y, Lysholm J.** Rating systems in the evaluation of knee ligament injuries. *Clin Orthop.* 1985;198:43-9.
38. **Beynnon BD, Risberg MA, Tjomsland O, Ekeland A, Fleming BC, Peura GD, Johnson RJ.** Evaluation of knee joint laxity and the structural properties of the anterior cruciate ligament graft in the human. A case report. *Am J Sports Med.* 1997;25:203-6.
39. **Blickenstaff KR, Grana WA, Egle D.** Analysis of a semitendinosus autograft in a rabbit model. *Am J Sports Med.* 1997;25:554-9.
40. **Grana WA, Egle DM, Mahnken R, Goodhart CW.** An analysis of autograft fixation after anterior cruciate ligament reconstruction in a rabbit model. *Am J Sports Med.* 1994;22:344-51.
41. **Liu SH, Panossian V, al-Shaikh R, Tomin E, Shepherd E, Finerman GA, Lane JM.** Morphology and matrix composition during early tendon to bone healing. *Clin Orthop.* 1997;339:253-60.
42. **Nagano M, Yoshiya S, Kuroda R, Kurosaka M, Mizuno K.** Remodeling and healing process of bone-patellar tendon-bone graft in a bone tunnel: a histological study in dogs. *Trans Orthop Res Soc.* 1997;22:78.
43. **Pinczewski LA, Clingeleffer AJ, Otto DD, Bonar SF, Corry IS.** Integration of hamstring tendon graft with bone in reconstruction of the anterior cruciate ligament. *Arthroscopy.* 1997;13:641-3.
44. **Rodeo SA, Arnoczky SP, Torzilli PA, Hidaka C, Warren RF.** Tendon-healing in a bone tunnel. A biomechanical and histological study in the dog. *J Bone Joint Surg Am.* 1993;75:1795-803.
45. **Scranton PE Jr, Lanzer WL, Ferguson MS, Kirkman TR, Pflaster DS.** Mechanisms of anterior cruciate ligament neovascularization and ligamentization. *Arthroscopy.* 1998;14:702-16.
46. **Hamner DL, Brown CH Jr, Steiner ME, Hecker AT, Hayes WC.** Hamstrings tendon grafts for reconstruction of the anterior cruciate ligament: biomechanical evaluation of multiple strands and tensioning techniques. *J Bone Joint Surg Am.* 1999;81:549-57.